

Padhmavathy Cebolu Srinivasan

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EDUCATION

San Jose State University
Master of Science: Data Analytics

San Jose, CA
Jan. 2023 – Dec 2024

Alagappa University
Master of Business Administration: Human Resource Management

Chennai, India
Aug. 2019 – May 2021

Sathyabama University
Bachelor of Science: Computer Science Engineering

Chennai, India
Aug. 2013 – Mar 2017

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, HTML/CSS, Bash/Linux

Cloud Platforms: AWS (S3, EC2, RDS), GCP (BigQuery, Compute Engine), Azure (Data Services)

DevOps & Tools: Jenkins, CI/CD, UDeploy, GitHub, Confluence, JIRA, Excel, Docker, MongoDB

Visualization & Monitoring: Tableau, Splunk, PowerBI

Libraries & Frameworks: Pandas, NumPy, Spark, Scikit-learn, PyTorch, TensorFlow, HuggingFace Transformers, NLP, LLMs, RAG

EXPERIENCE

Data Scientist
CDUS Trading LLC

May. 2024 – Current
Chicago, IL

- Led stock market trend analysis by integrating technical indicators, financial reports, and historical data, leveraging key performance indicators (KPIs) for actionable insights.
- Designed and deployed a **sentiment-driven SPX market bias forecasting system** using **NLP** and **OpenAI**, reducing decision latency by 50% through automated pre-market analysis.
- Built and validated an **XGBoost**-based machine learning model to predict S&P 500 market direction, integrating technical indicators, sentiment scores, and futures data for improved forecasting accuracy.
- Enhanced decision-making speed by 40% through rigorous backtesting, model performance validation, and strategic documentation; created **Tableau dashboards** to visualize trends and model outcomes.

Business Data Analyst
Capgemini, Synchrony Bank (Client)

June 2017 – Nov 2020
Chennai, India

- Engineered distributed log analysis tools using SQL and Python, cutting incident resolution time by 35%.
- Enhanced system reliability by 40%, automating cloud service restarts and auto-scaling using **Python scripts**, minimizing manual interventions.
- Actively collaborated with cross-functional teams in Scrum calls to align on business goals, track sprint progress, and enhance operational efficiency.
- Built NLP-driven log parsers to automatically extract error patterns and critical event keywords from system logs, improving root cause analysis speed and accuracy.
- Designed and maintained **Splunk and Tableau dashboards** to monitor cloud service performance, enhancing real-time visibility into system health and supporting data-driven decision-making.

PROJECTS

Docubot | *Python, RAG, LLaMA-2, ChromaDB, openai, NLP*

Developed and implemented Retrieval-Augmented Generation (RAG) using LLaMA-2 (7B), integrating ChromaDB and LangChain for optimized contextual responses. Designed an efficient pipeline leveraging quantized models and OpenAI API for enhanced retrieval and generation. [Github](#)

Smart City AI Traffic Cloud - City Traffic Analysis and Monitoring using AI Models | *Python, ML & AI, Yolo, DeepSORT, DCRNN, ReactJS, MongoDB*

Developed CCTV-based traffic flow and accident detection models using YOLOv8 for object detection. Integrated drone-based vehicle detection and tracking using YOLOv8 + DeepSORT for real-time traffic analysis. Built a multi-cluster DCRNN model for traffic speed and flow forecasting using IoT sensor data, enabling predictive analysis. (**Research paper accepted at IEEE CAI 2025 (Conference on Artificial Intelligence)**) [Github-Smartcity](#)

Market Alert Bot | *Python, OpenAI API, Logistic Regression, Github Actions*

Developed an automated SPX market forecasting tool using Python, OpenAI, and logistic regression to analyze news sentiment, futures, and volatility data, with daily market alert emails sent via GitHub Actions.